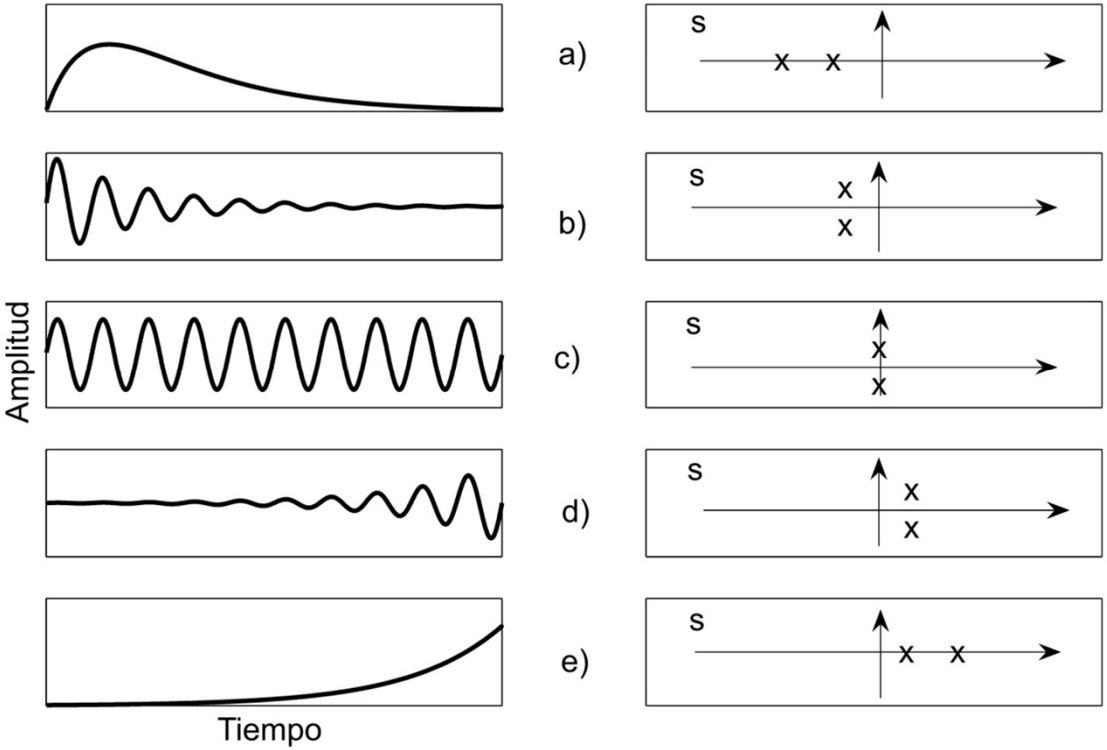
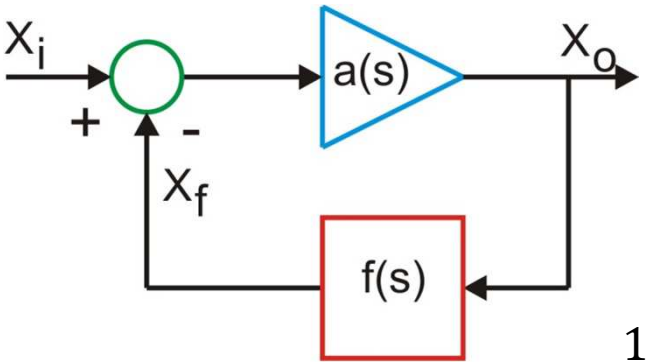
Principios básicos de Osciladores Senoidales



Principios básicos de Osciladores Senoidales

- ✓ Osciladores lineales en sentido matemático.
- ✓ La variación de los parámetros y la dependencia de las condiciones iniciales, hacen necesario un control no lineal de la ganancia.
- ✓ Para el análisis y diseño de estos circuitos se utilizan, en principio, los métodos lineales de circuitos realimentados.
- ✓ A continuación se puede obtener un mecanismo no lineal para el control de la amplitud.

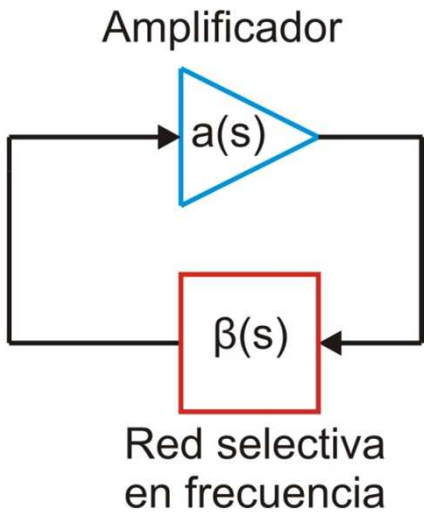
Criterio de oscilación



$$A(s) = \frac{X_o}{X_i} = \frac{a(s)}{1 + a(s)f(s)}$$

$$T(s) = a(s)f(s)$$

$$1 + T(s) = 0 \quad \rightarrow \quad T(s) = -1 \;\; oscilaciones\ sostenidas$$



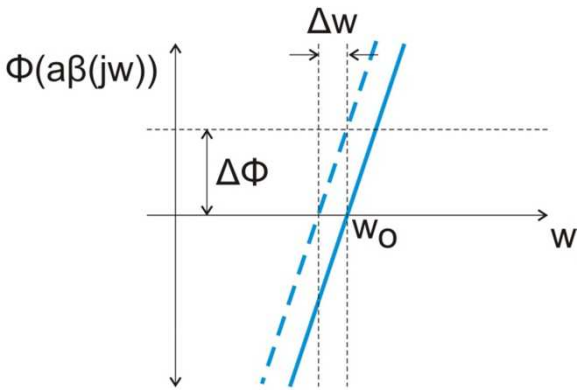
$$Criterio\ de\ Barkhausen$$

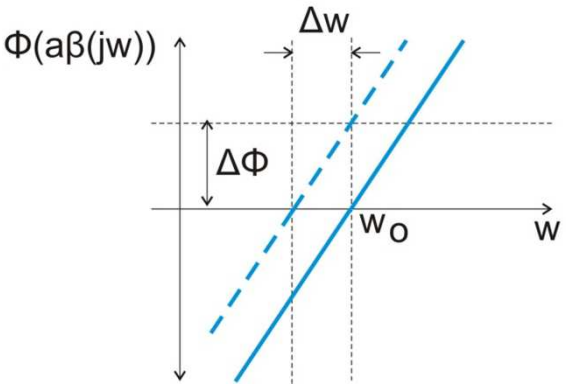
$$a(s)\beta(s) = 1$$

$$a\beta(jw_o) = 1 + j0 \;\; ; \;\; w_o \;\; frecuencia\ de\ oscilación$$

$$\begin{cases} \emptyset(a\beta(jw_o)) = 0 & (condición\ de\ fase) \\ |a\beta(jw_o)| = 1 & (condición\ de\ amplitud) \end{cases}$$

Estabilidad en frecuencia





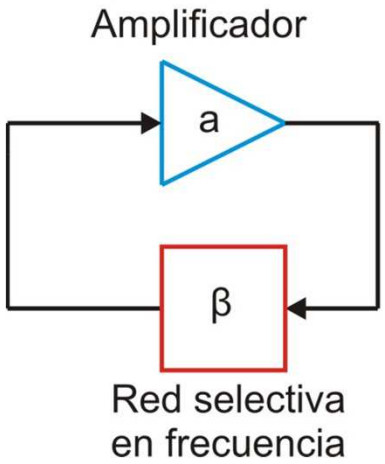
Factor de estabilidad en frecuencia

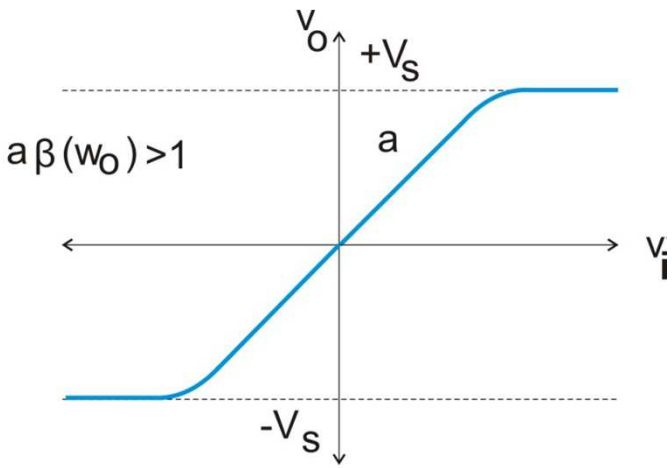
$$SF = w_o \left| \frac{\partial \emptyset}{\partial w} \right|_{w=w_o}$$

Control de Amplitud

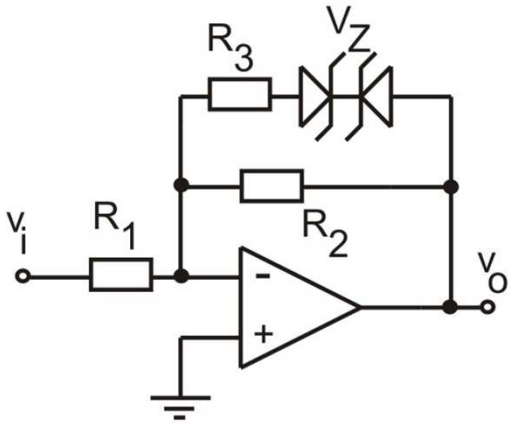
El criterio de Barkhausen garantiza oscilaciones senoidales en un sentido matemático. Criterio modificado:

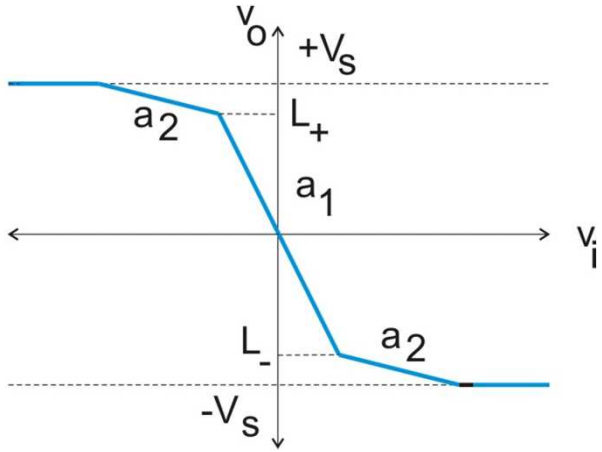
$$\begin{cases} \emptyset(a\beta(jw_o)) = 0 \\ |a\beta(jw_o)| > 1 \end{cases}$$





Control de Amplitud



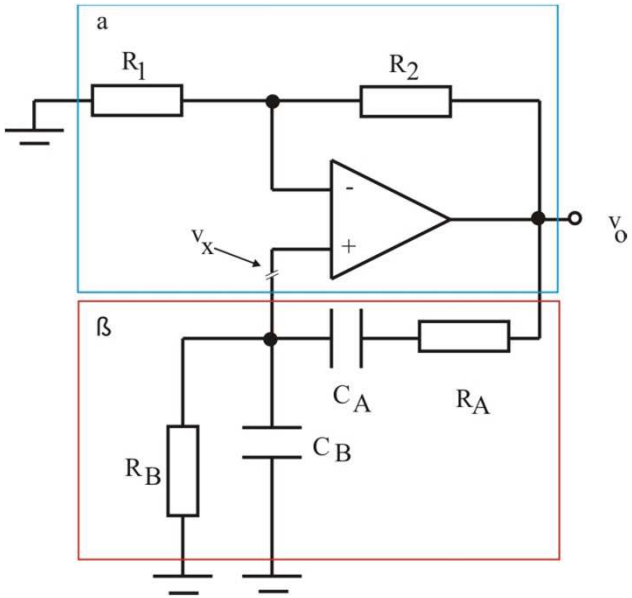


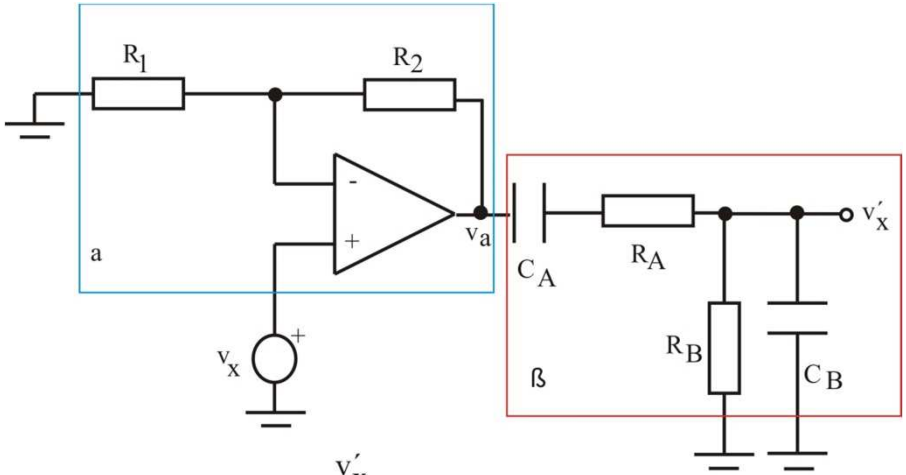
$$a_1 = -\frac{R_2}{R_1} \quad ; \qquad |a_1\beta(w_o)| > 1$$

$$L_+ = -L_- = V_Z + V_D$$

$$a_2 = -\frac{R_2||R_3}{R_1} \quad ; \qquad |a_2\beta(w_o)| < 1$$

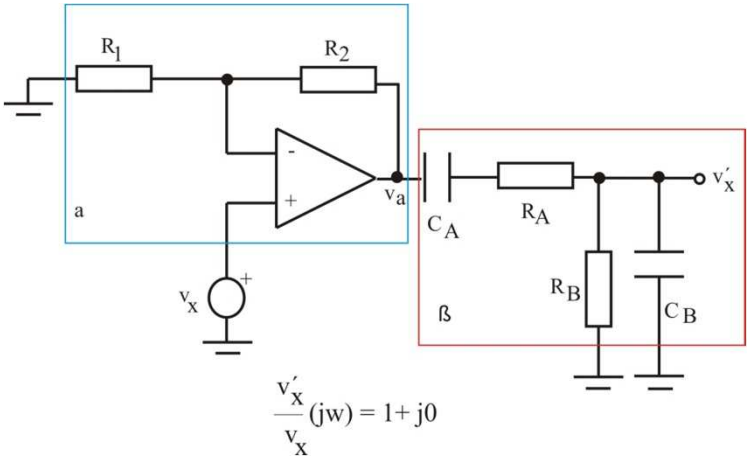
Oscilador puente de Wien





$$\frac{v'_x}{v_x}(jw) = 1 + j0$$

Oscilador puente de Wien



$$a\beta(jw) = \frac{v_x{}'}{v_x} = \frac{v_x{}'}{v_a}\frac{v_a}{v_x} = \frac{\dfrac{R_B}{1+jwC_BR_B}}{\dfrac{R_B}{1+jwC_BR_B} + R_A + \dfrac{1}{jwC_A}}\left(1+\frac{R_2}{R_1}\right)$$

$$= \frac{jwC_AR_B}{1 - w^2C_AC_BR_AR_B + jw(C_AR_B + C_AR_A + C_BR_B)}\left(1+\frac{R_1}{R_2}\right)$$

Oscilador puente de Wien

$$a\beta(jw) = \frac{jwC_AR_B}{1 - w^2C_AC_BR_AR_B + jw(C_AR_B + C_AR_A + C_BR_B)}\left(1 + \frac{R_2}{R_1}\right) = 1 + j0$$

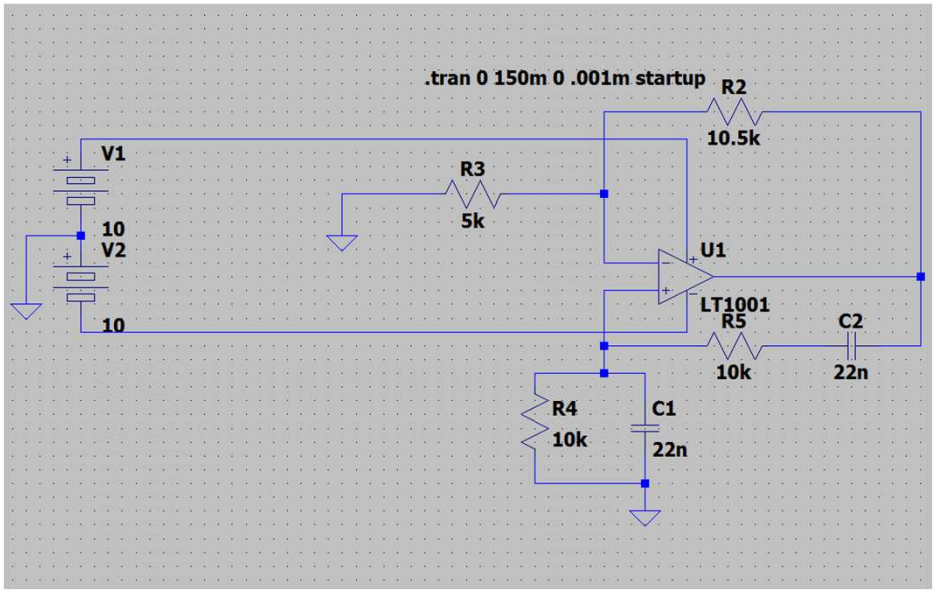
Condición de fase:

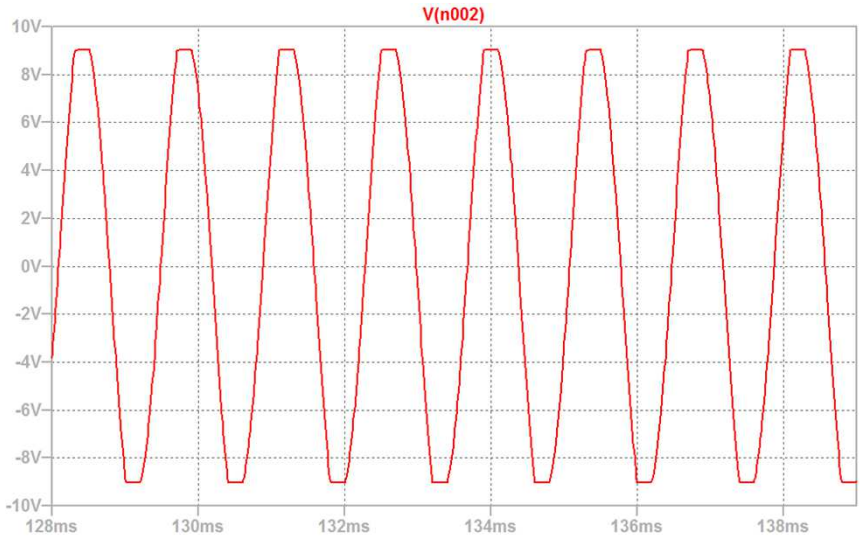
$$1 - w^2C_AC_BR_AR_B = 0 \quad \rightarrow \quad w_0 = \frac{1}{\sqrt{C_AC_BR_AR_B}}$$

Condición de amplitud:

$$\frac{\left(1 + \frac{R_2}{R_1}\right)C_AR_B}{(C_AR_B + C_AR_A + C_BR_B)} > 1 \qquad \rightarrow \left(1 + \frac{R_2}{R_1}\right) > 1 + \frac{R_A}{R_B} + \frac{C_B}{C_A}$$

Oscilador puente de Wien

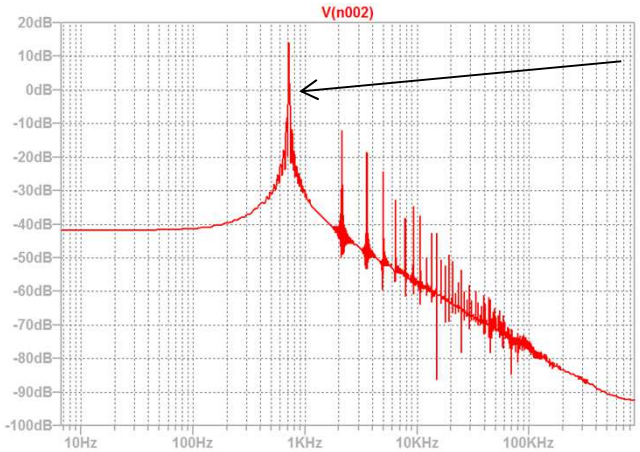




$$R_A = R_B = 10K \;\; ; \;\; C_A = C_B = 22n$$

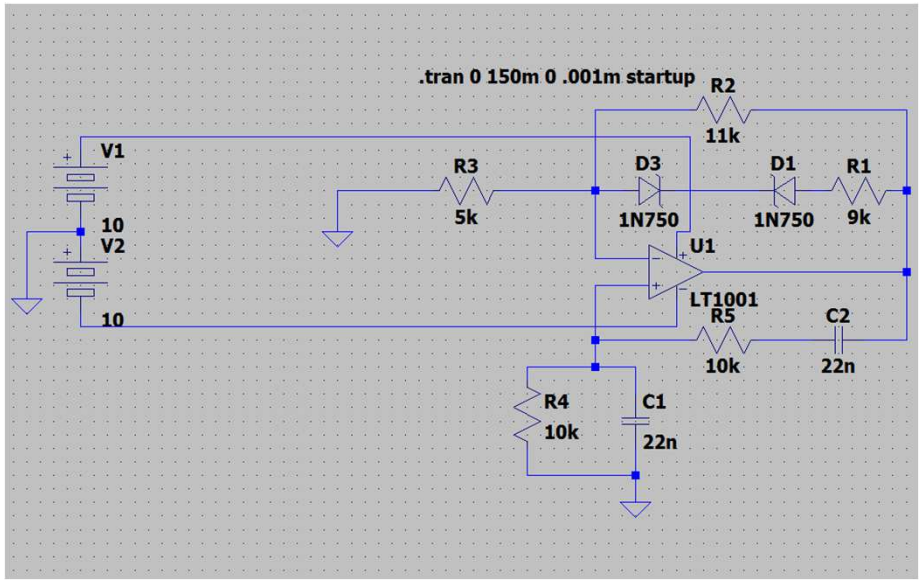
$$f_o = \frac{1}{2\pi 10K22n} = 723{,}4\ Hz$$

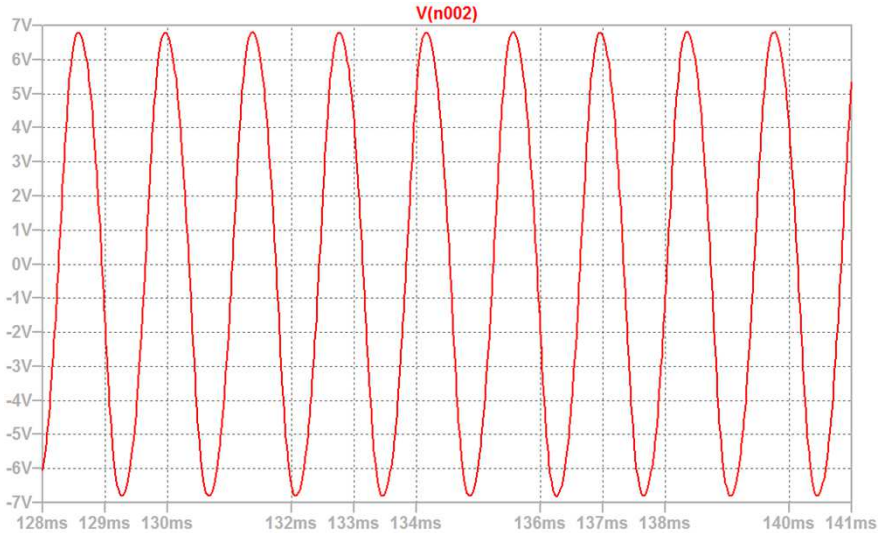
$$1 + \frac{R_2}{R_3} > 3$$



$713{,}3\ Hz$

Oscilador puente de Wien





$$R_A = R_B = 10K \;\; ; \;\; C_A = C_B = 22n$$

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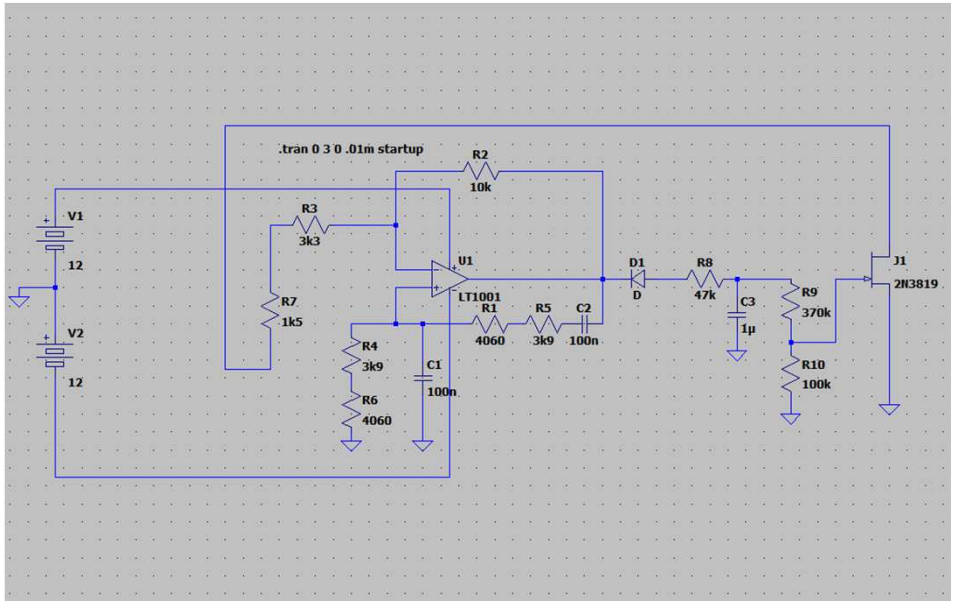
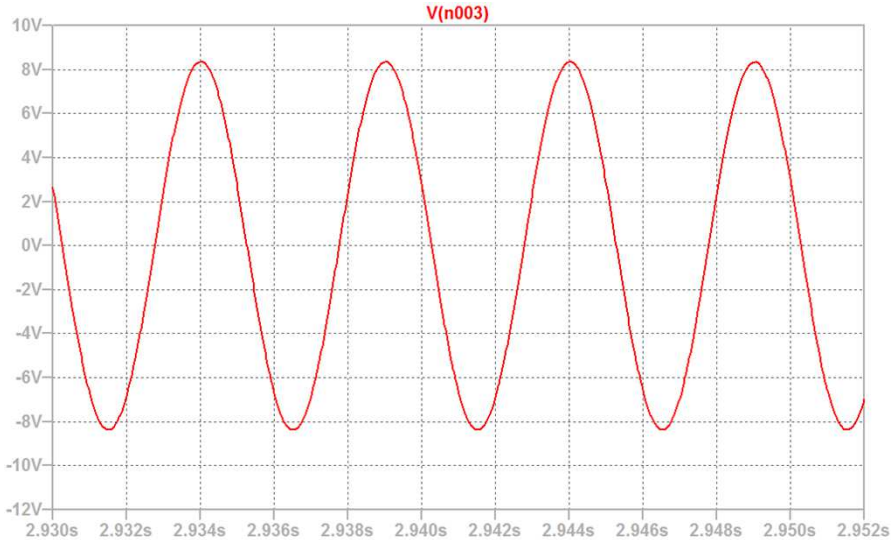
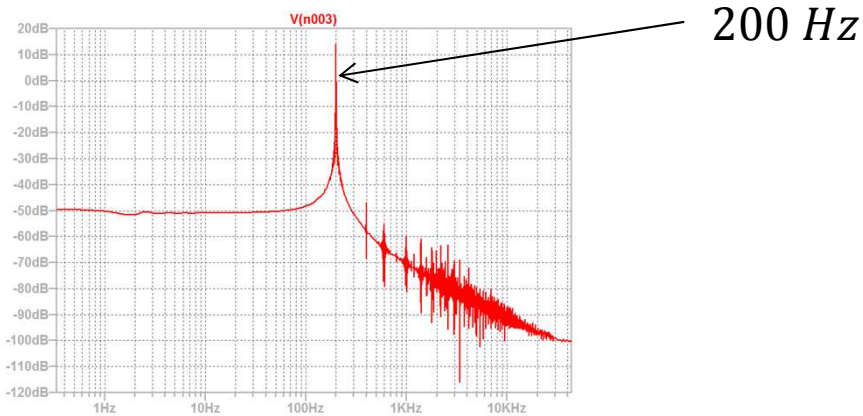
Oscilador puente de Wien



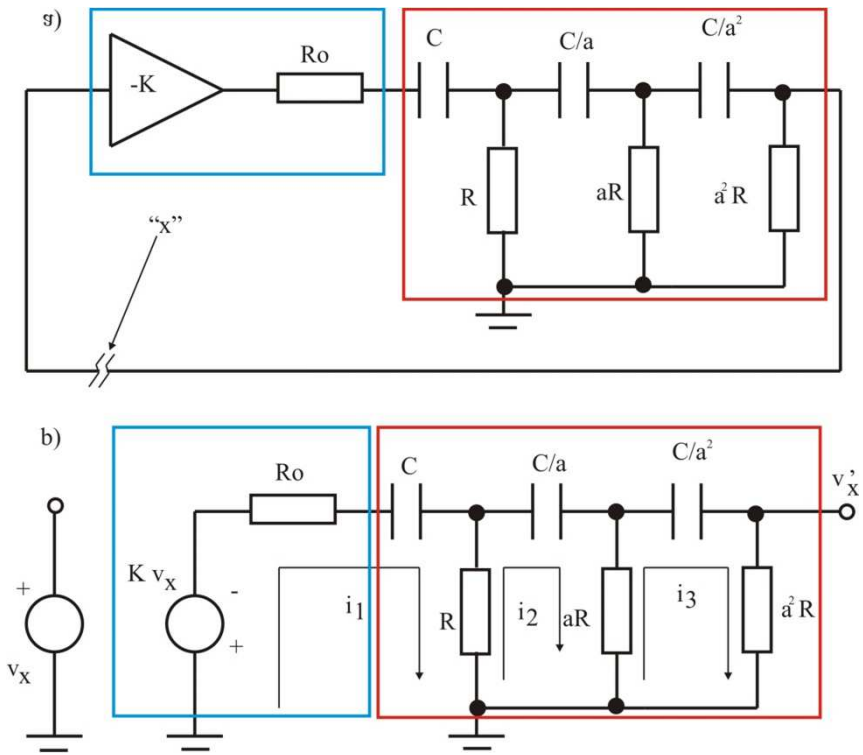


$$R_A = R_B = 7{,}96K \;\; ; \;\; C_A = C_B = 100n$$

$$f_o = \frac{1}{2\pi 7{,}96K100n} = 199{,}9Hz$$



Oscilador de deplazamiento de fase



$$\begin{cases} -Kv_x = i_1\left(R_0 + R + \dfrac{1}{jwC}\right) - i_2R \\ 0 = -i_1R + i_2\left(R + aR + \dfrac{a}{jwC}\right) - i_3aR \\ 0 = -i_2aR + i_3\left(aR + a^2R + \dfrac{a^2}{jwC}\right) \end{cases}$$

$$v_{x'} = i_3a^2R$$

Oscilador de desplazamiento de fase

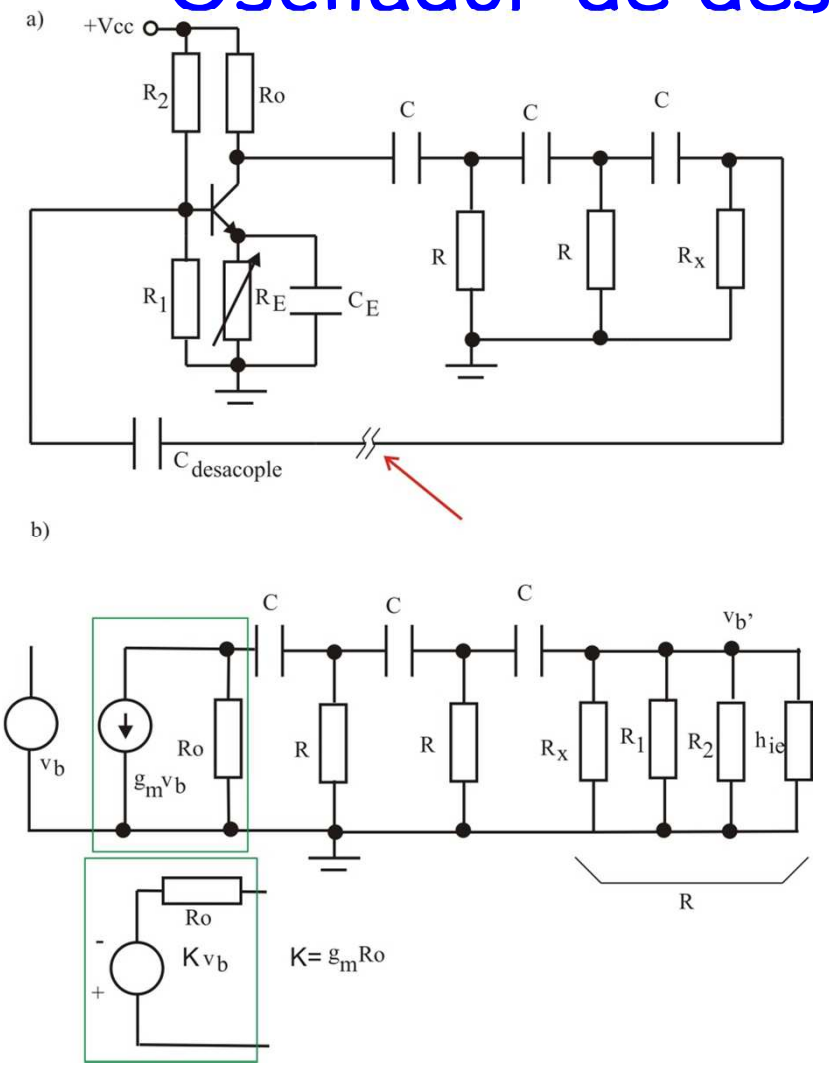
De las expresiones anteriores se halla la ganancia de lazo $\frac{v_{x'}}{v_x}(jw)$
Si se anula la parte imaginaria, condición de fase, se obtiene:

$$w_0 = \frac{1}{RC\sqrt{3 + \frac{2}{a} + \frac{1}{a^2} + \frac{R_0}{R}\left(\frac{2}{a} + 2\right)}}$$

Si se cumple que a la frecuencia w_0 la transferencia $\frac{v_{x'}}{v_x}(jw_0)$,
(condición de amplitud), el circuito oscilará. De esta condición:

$$K > 8 + \frac{12}{a} + \frac{7}{a^2} + \frac{2}{a^3} + \frac{R_0}{R}\left(8 + \frac{11}{a} + \frac{4}{a^2}\right) + \left(\frac{R_0}{R}\right)^2\left(2 + \frac{2}{a}\right)$$

Oscilador de desplazamiento de fase

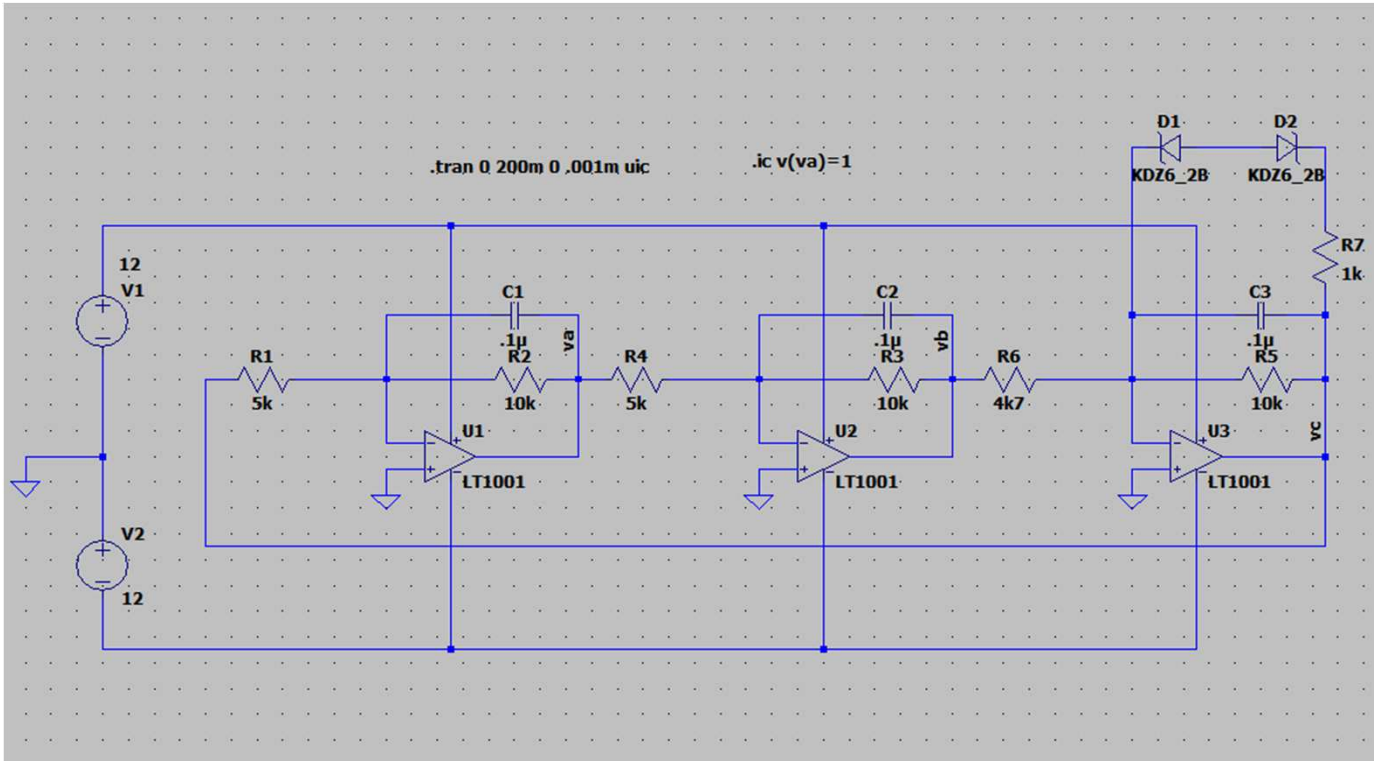


$$a = 1 \; ; \quad R_x \| R_1 \| R_2 \| h_{ie} \quad ; \quad K = g_m R_0$$

$$f_{osc} = \frac{1}{2\pi\sqrt{6 + 4\frac{R_0}{R}}}$$

$$K > 29 + 23\frac{R_0}{R} + 4\left(\frac{R_0}{R}\right)^2$$

Oscilador Trifásico



$$R_2 = R_3 = R_5 = R_A \;\; ; \;\; C_1 = C_2 = C_3 = C_A \;\; ; \;\; R_1 = R_4 = R_6 = R_B$$

$$Tres\ etapas\ idénticas : -\frac{R_A}{R_B}\frac{1}{(1+jwR_AC_A)} \quad \rightarrow a\beta(jw) = -\left(\frac{R_A}{R_B}\right)^3\left(\frac{1}{1+jwR_AC_A}\right)^3$$

Oscilador Trifásico

$$a\beta(jw) = \left(-\frac{R_A}{R_B}\frac{1}{(1+jwR_AC_A)}\right)^3 = -\left(\frac{R_A}{R_B}\right)^3\left(\frac{1}{1-3w^2{C_A}^2{R_A}^2+j\left(3wC_AR_A-w^3{C_A}^3{R_A}^3\right)}\right)$$

$$a\beta(jw_0) = 1 + j0$$

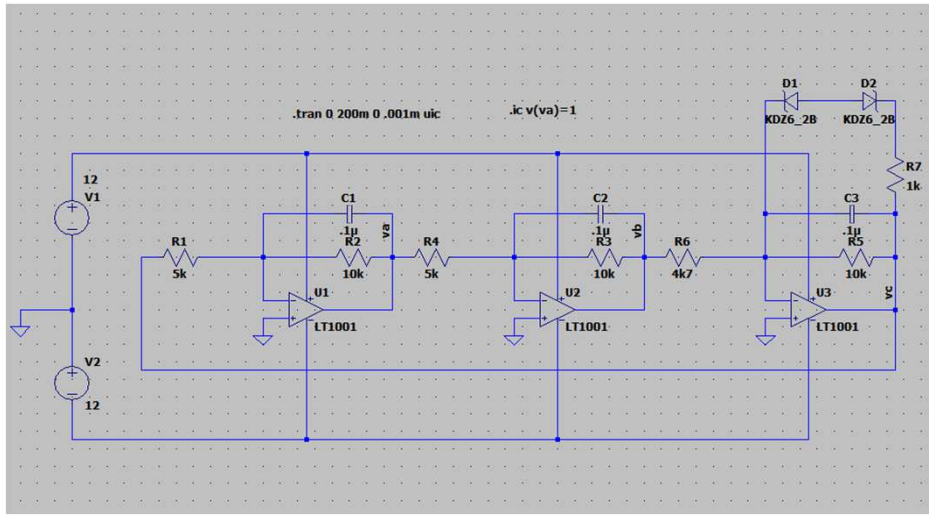
$$3w_0C_AR_A - {w_0}^3{C_A}^3{R_A}^3 = 0 \;\; ; \quad Condición\ de\ fase$$

$$\boxed{f_{osc} = \frac{\sqrt{3}}{2\pi C_AR_A}}$$

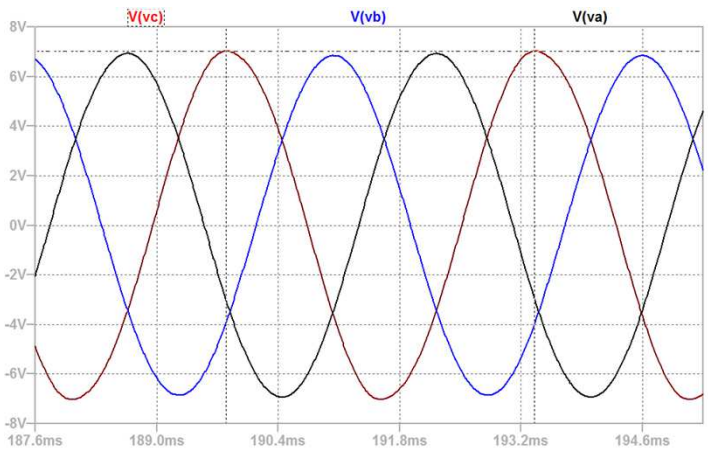
$$-\left(\frac{R_A}{R_B}\right)^3\frac{1}{1-3{w_0}^2{C_A}^2{R_A}^2} = -\left(\frac{R_A}{R_B}\right)^3\frac{1}{1-3*3} \geq 1 \;\; ; \quad Condición\ de\ amplitud$$

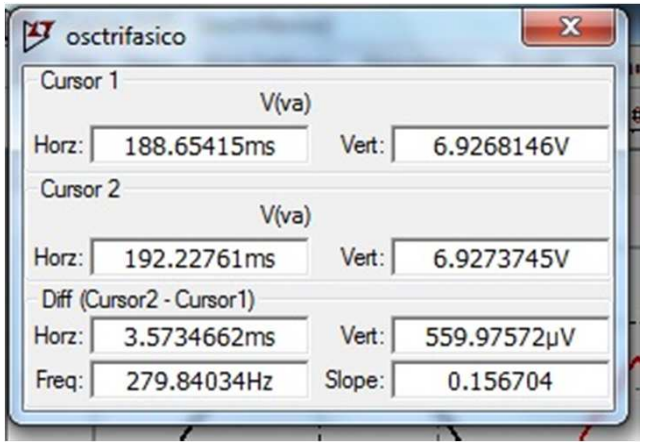
$$\boxed{\left(\frac{R_A}{R_B}\right) \geq 2}$$

Oscilador Trifásico

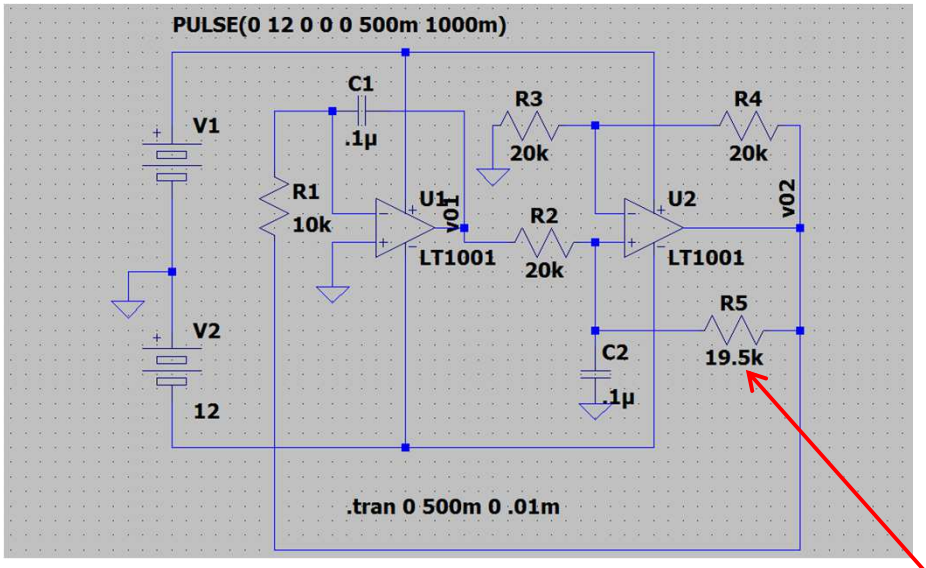


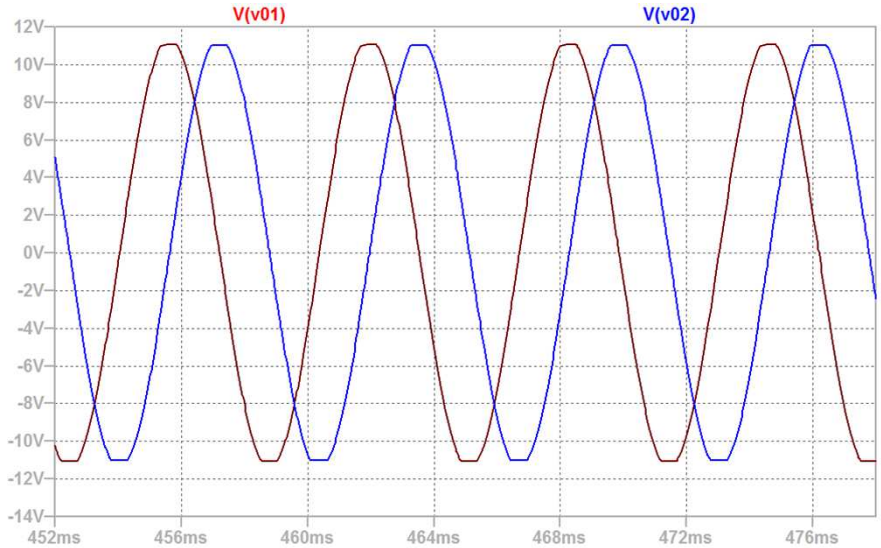
$$f_{osc} = \frac{\sqrt{3}}{2\pi R_A C_A} = 275{,}7Hz$$





Oscilador de cuadratura





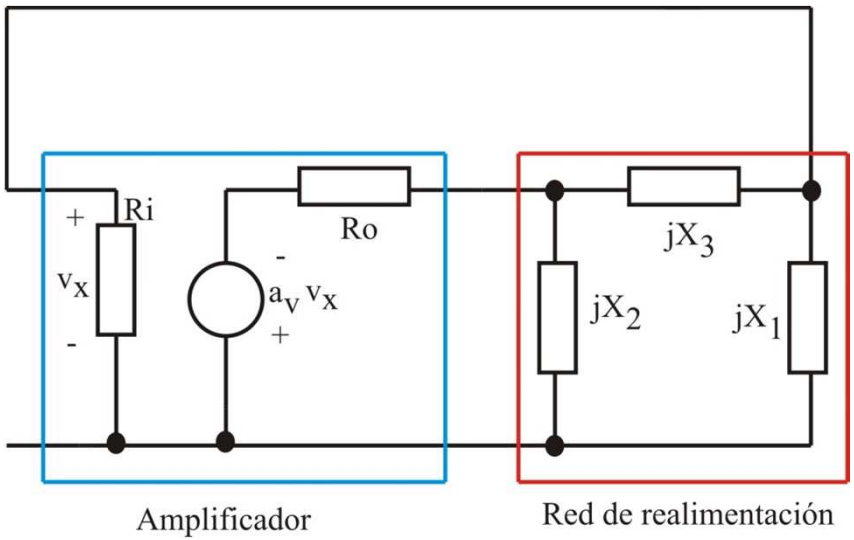
$< 2R$

$$a\beta(s) = -\frac{1}{s^2R^2C^2} \quad ; \quad R = 10K \quad C = 0{,}1\mu F$$

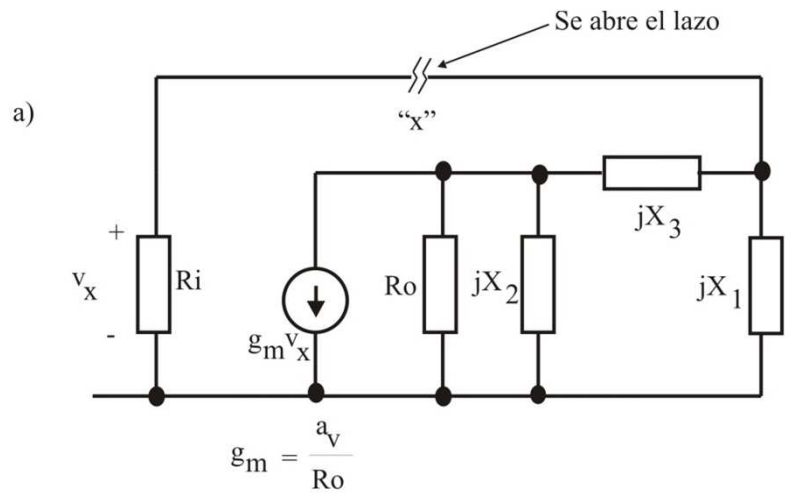
$$a\beta(jw) = \frac{1}{jw^2R^2c^2} \quad ; \quad f_{osc} = \frac{1}{2\pi RC} \cong 159{,}1Hz$$

Oscilador L-C sintonizado

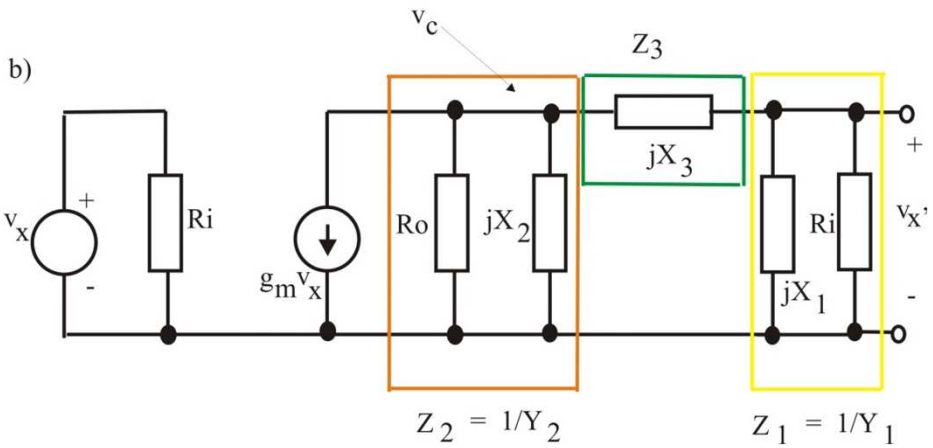
- ✓ Se usan a frecuencias superiores que los R-C (100KHz a cientos de MHz)
- ✓ Gran selectividad en circuitos L-C con alto factor de calidad Q
- ✓ Buena estabilidad en frecuencia: $SF \cong 2 * Q$



Oscilador L-C sintonizado



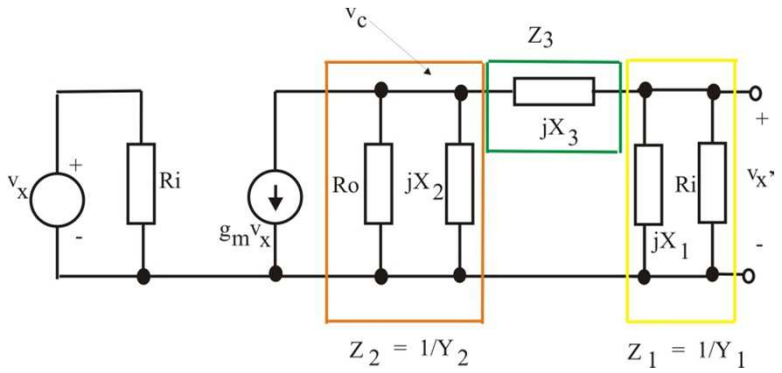
$$\begin{cases} v_c Y_2 + g_m v_x + \dfrac{(v_c - v_{x'})}{Z_3} = 0 \\ v_{x'} Y_1 + \dfrac{(v_{x'} - v_c)}{Z_3} = 0 \end{cases}$$



Criterio de oscilación

$$\boxed{\frac{v_{x'}}{v_x}(jw_0) = \frac{-g_m}{Y_1 + Y_2 + Y_1 Y_2 Z_3} = 1 + j0}$$

Oscilador L-C sintonizado



$$R_i \gg |X_1|_{w_0} \rightarrow Y_1 = \frac{1}{jX_1}\ ,\ \ Z_3 = jX_3 \ \ e \ \ \ Y_2 = \frac{1}{R_0} + \frac{1}{jX_2}$$

$$\frac{v_{x'}}{v_x}(jw_0) = \frac{-g_m}{Y_1 + Y_2 + Y_1 Y_2 Z_3} = 1 + j0$$

$$\frac{v_{x'}}{v_x}(jw_0) = \frac{-g_m}{\frac{1}{jX_1} + \frac{1}{R_0} + \frac{1}{jX_2} + \frac{jX_3}{jX_1}\left(\frac{1}{R_0} + \frac{1}{jX_2}\right)} = \frac{-g_m}{\frac{1}{R_0} + \frac{X_3}{X_1 R_0} + \frac{1}{j}\underbrace{\left(\frac{X_1 + X_2 + X_3}{X_1 X_2}\right)}_{=0}} = 1 + j0$$

$$\frac{-g_m}{\frac{1}{R_0}\left(1 + \frac{X_3}{X_1}\right)} = \frac{-g_m R_0 X_1}{X_1 + X_3} = \frac{-g_m R_0 X_1}{-X_2} = 1$$

$$\boxed{X_1 + X_2 + X_3 = 0}$$

$$\boxed{g_m R_0 > \frac{X_2}{X_1}}$$

Condición de amplitud

Condición de fase

Oscilador L-C sintonizado

Condición de fase

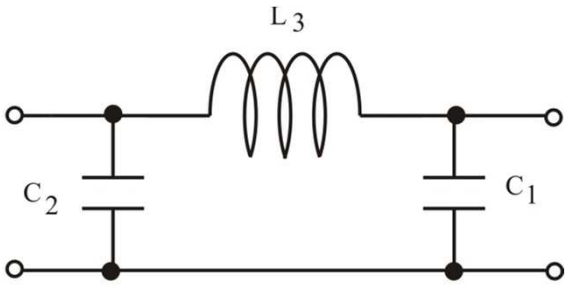
$$X_1 + X_2 + X_3 = 0$$

Condición de amplitud

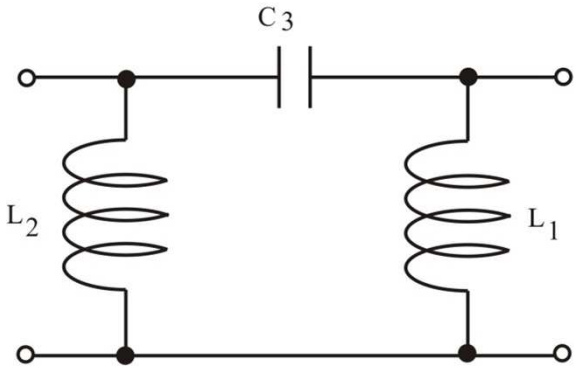
$$g_m R_0 > \frac{X_2}{X_1}$$

$$signo\ X_1 = signo\ X_2$$

$$signo\ X_3 \neq signo X_{1-2}$$

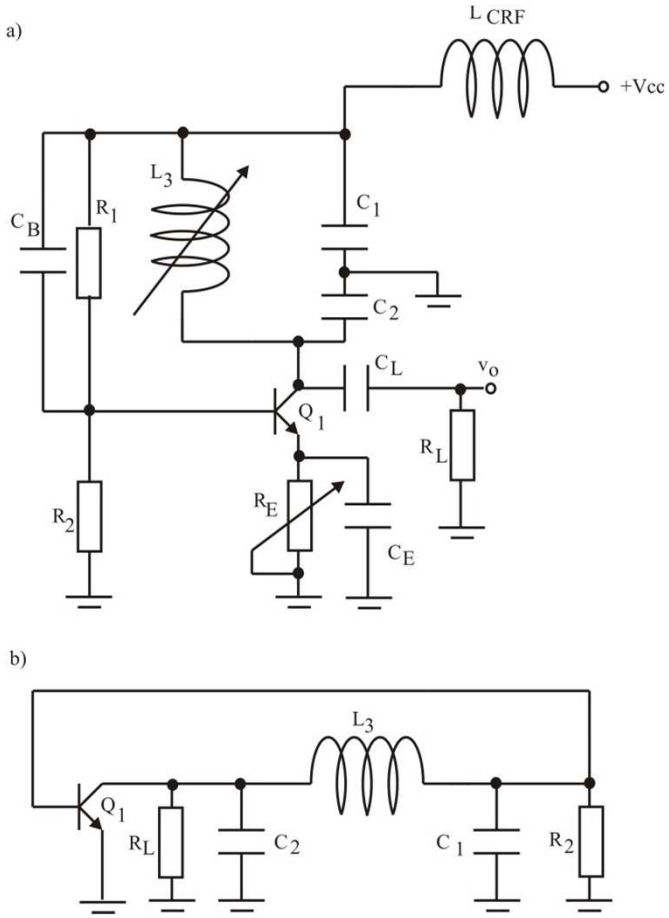


Colpitts



Hartley

Oscilador Colpitts

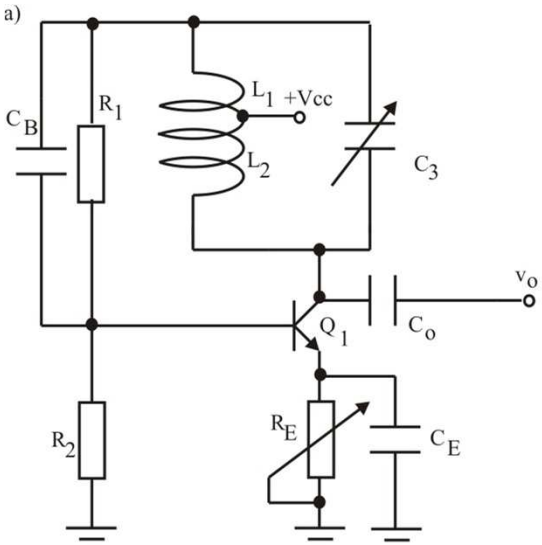


$$-\frac{1}{w_0 C_1} - \frac{1}{w_0 C_2} + w_0 L_3 = 0$$

$$\boxed{w_0 = \frac{1}{\sqrt{L_3 \frac{C_1 C_2}{C_1 + C_2}}}}$$

$$\boxed{g_m R_L > \frac{X_2}{X_1} = \frac{C_1}{C_2}}$$

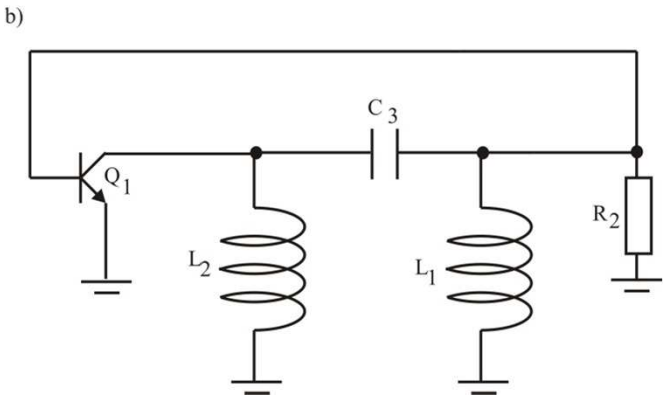
Oscilador Hartley



$$w_0 L_1 + w_0 L_2 - \frac{1}{w_0 C_3} = 0$$

$$\boxed{w_0 = \frac{1}{\sqrt{(L_1 + L_2)C_3}}}$$

$$\boxed{g_m R_0 > \frac{X_2}{X_1} = \frac{L_2}{L_1}}$$

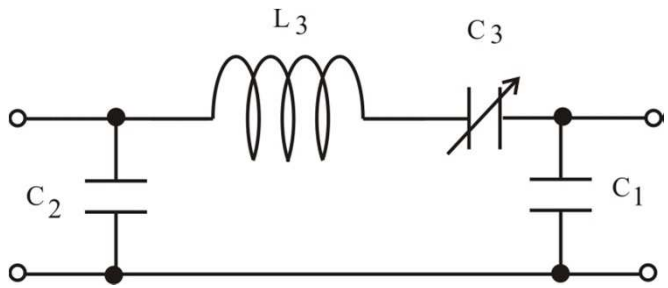


Con inductancia mutua entre L_1 y L_2

$$\boxed{w_0 = \frac{1}{\sqrt{(L_1 + L_2 + 2M)C_3}}}$$

$$\boxed{g_m R_0 > \frac{X_2}{X_1} = \frac{L_2 + M}{L_1 + M}}$$

Oscilador Clapp



$$-\frac{1}{w_0 C_1} - \frac{1}{w_0 C_2} - \frac{1}{w_0 C_3} + w_0 L_3 = 0$$

$$w_0 = \frac{1}{\sqrt{L_3 \frac{1}{\frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}}}}$$

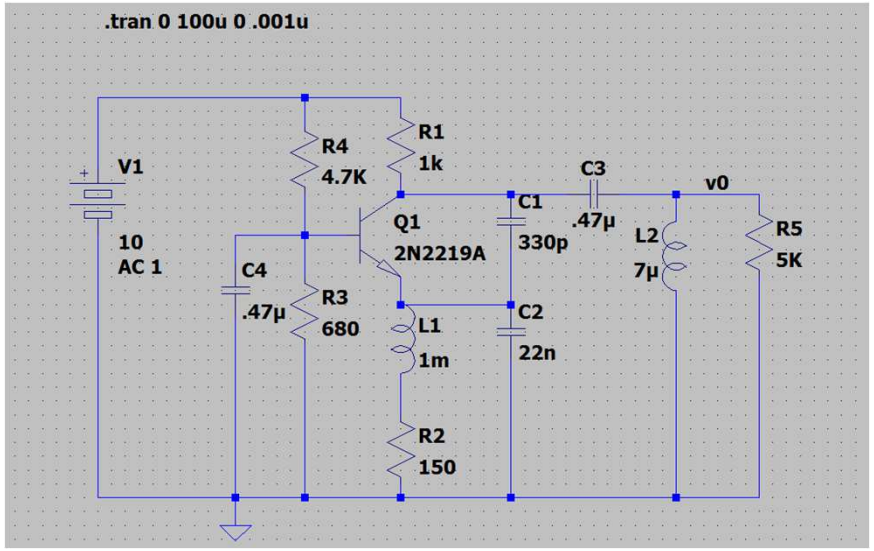
$$Si\ C_3 \ll C_1, C_2 \rightarrow$$

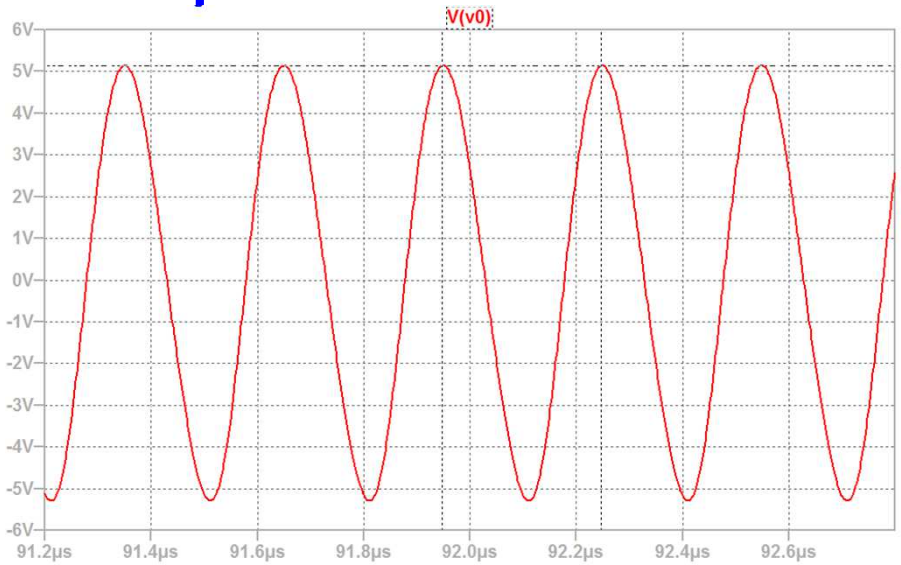
$$w_o \cong \frac{1}{\sqrt{L_3 C_3}}$$

$$g_m R_0 > \frac{X_2}{X_1} = \frac{C_1}{C_2}$$

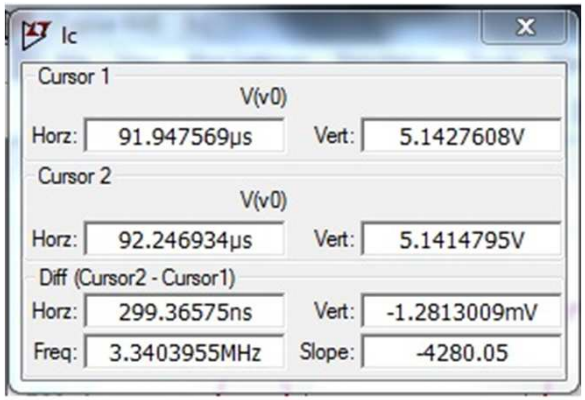
Se puede ajustar la frecuencia mediante C_3, independientemente de la condición de amplitud.

Oscilador Colpitts



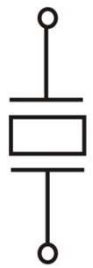


$$f_{osc} = \frac{1}{2\pi\sqrt{7\mu H \frac{22000pF\ 330pF}{22330pF}}} = 3{,}336MHz$$

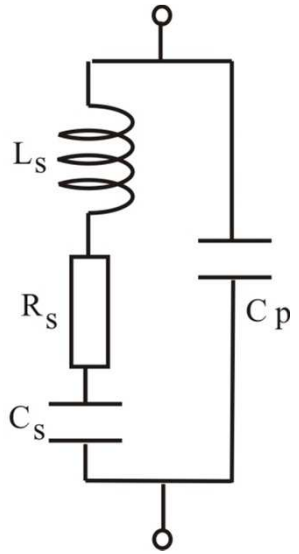


Oscilador de cristal

- ✓ Un cristal piezoeléctrico de cuarzo, posee características de resonancia electromecánica muy estables (con tiempo y temperatura) y altamente selectivas (factores de calidad Q muy altos)



Símbolo



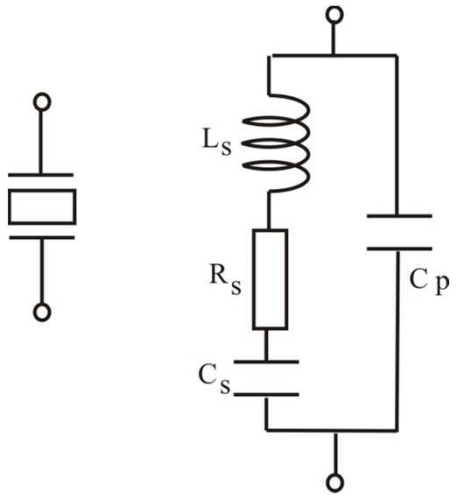
Circuito equivalente

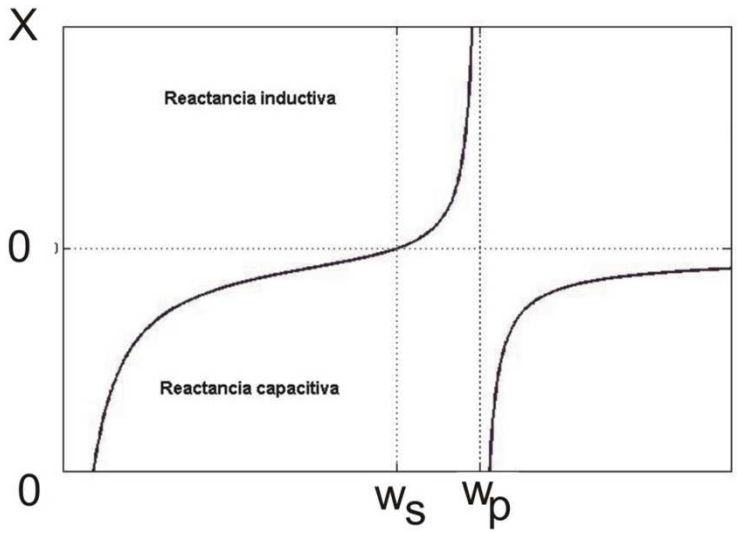
```
* 1Mhz frequency standard, AT cut, parallel
* resonant, Q=25000, calibration capacitance = 13pf

.subckt QZP1MEG 1 2
  Ls    1    11    2.54647909
  Cs    11   12    9.95357648e-015
  Rs    12   2     640
  Cp    1    2     2.48839412e-012
.ends

*$
```

Oscilador de cristal





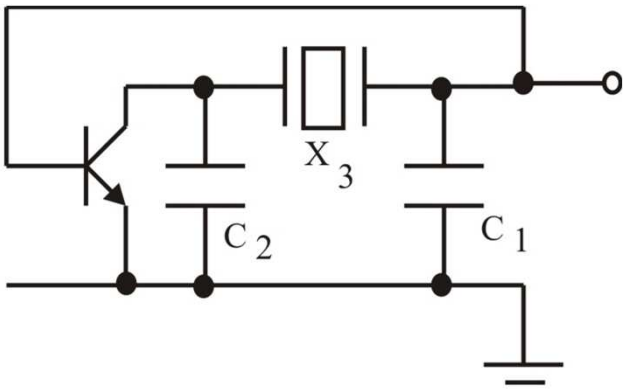
Debido al muy alto Q, Rs puede ser despreciada

$$Z_{xtal} = jX = \frac{-j}{wC_p}\left(\frac{w^2 - {w_s}^2}{w^2 - {w_p}^2}\right)$$

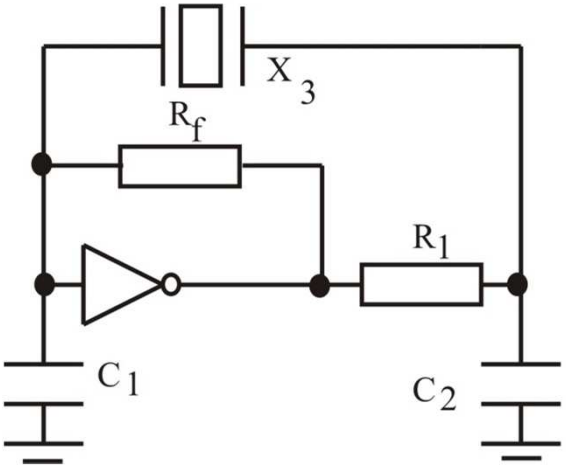
$$w_s = \frac{1}{\sqrt{L_s C_s}} \quad y \quad w_p = \frac{1}{\sqrt{L_s \frac{1}{\frac{1}{C_s} + \frac{1}{C_p}}}}$$

$$como \;\; C_p \gg C_s \;\; \rightarrow \;\; w_s \;\; y \;\; w_p \;\; difieren\ muy\ poco\ numéricamente$$

Oscilador Pierce

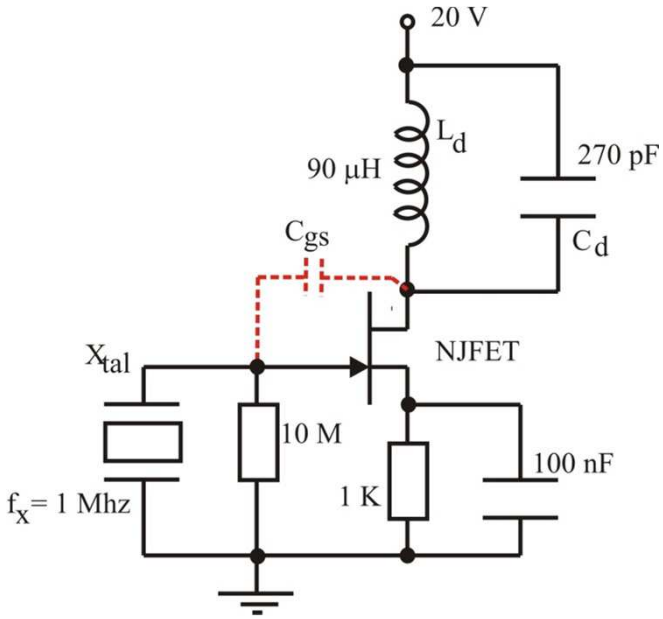


- ✓ Se reemplaza L3 en un Colpitts por el Cristal
- ✓ Existe siempre un valor de frecuencia entre ws y wp tal que se cumple: X1+X2+X3=0
- ✓ Se debe asegurar la condición de amplitud para obtener oscilaciones sostenidas



- ✓ Versión del oscilador Pierce con inversor CMOS

Oscilador de cristal



- ✓ Se reemplaza L1 en un Hartley por el Cristal
- ✓ Existe siempre un valor de frecuencia entre ws y wp tal que se cumple la condición de fase
- ✓ Se debe asegurar la condición de amplitud para obtener oscilaciones sostenidas

$$X_2 = X_d = \frac{wL_d}{1 - w^2 L_d C_d}$$

$$X_1 = X_{tal}$$

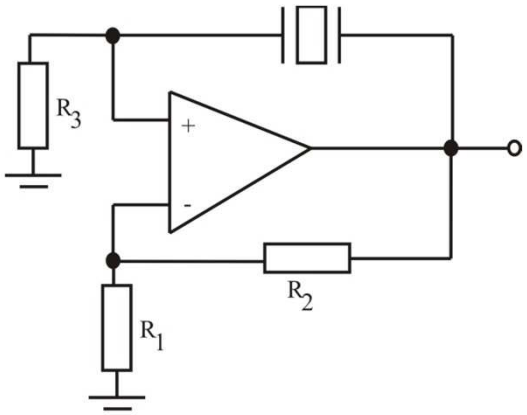
$$X_3 = -\frac{1}{wC_{gs}}$$

$$X_1 + X_2 + X_3 = 0$$

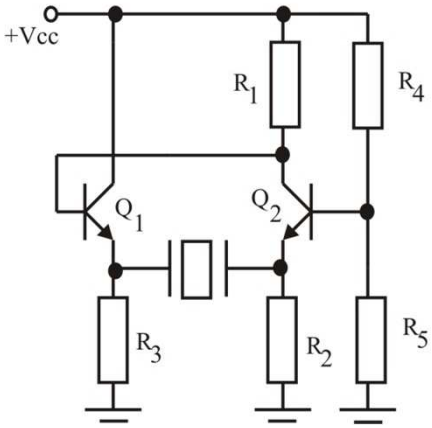
$$f_{osc} \cong 1MHz$$

$$Ganacia\ de\ JFET \quad g_m R_D = \mu > \frac{X_d}{X_{tal}}$$

Oscilador serie



$$1 + \frac{R_2}{R_1} > \frac{R_3}{R_3 + R_s} \quad condición\ de\ amplitud$$



- ✓ El cristal resuena a la frecuencia serie fs ofreciendo una impedancia resistiva pura Rs
- ✓ Se debe asegurar la condición de amplitud para obtener oscilaciones sostenidas